

Demo Cockpit Starter • Systems Reliability Blueprint

Distributed Ingestion Queues • PostgreSQL Query Acceleration • Connection Pooling • AI Gateway Defense

Client: Enterprise SaaS Founder / CTO

Engagement: Systems Reliability & Production Architecture

Calibrated Rate: \$40.00/hr USD (or Phased Turnkey)

Live Prototype: VERIFIED & OPERATIONAL

PRODUCTION SCOPE & MILESTONE DELIVERY SCHEDULE

Live Cockpit: <https://demo-cockpit-starter.vercel.app>

MILESTONE	ARCHITECTURE & PRODUCTION ENGINEERING DELIVERABLES	TIMELINE	SHARE	ALLOCATION
Phase 0	Interactive Architecture Prototype & Operational Cockpit (Live) Living demo: High-throughput ingestion buffer simulation, PostgreSQL query plan acceleration, connection pooling multiplexing, cache stampede defense, and sub-500ms AI gateway circuit breaker.	Live Now	Included	\$0.00 (Live)
Milestone 1	Codebase Discovery, Observability Baseline & Staging CI/CD Audit Audit current backend codebase (FastAPI/Django/Node), profile runtime latency, review Docker/K8s compose configurations, establish Prometheus/Datadog metrics baseline, and ship initial atomic PR to validate CI/CD pipeline.	Week 1	20%	25 Hrs (\$1,000)
Milestone 2	PostgreSQL Slow Query Profiling, Execution Plans & Composite Indexing Analyze EXPLAIN (ANALYZE, BUFFERS) telemetry on top bottleneck queries. Eliminate disk buffer sequential scans via composite covering B-Trees, implement zero-downtime CONCURRENT index migrations, and cut P99 DB latency by 90%+.	Week 2	20%	25 Hrs (\$1,000)
Milestone 3	Asynchronous Ingestion Queues, Redis Streams & Autoscaled Workers Decouple synchronous HTTP request threads with Redis 7 Streams buffer. Scale Celery/Inngest consumers under traffic surges, configure exponential backoff retries with full jitter, and wire Dead-Letter Queue (DLQ) alerts.	Week 3	20%	25 Hrs (\$1,000)
Milestone 4	PgBouncer Connection Pooling & Redis XFetch Stampede Elimination Deploy PgBouncer in transaction pooling mode multiplexing thousands of web connections to dedicated PostgreSQL server slots. Implement probabilistic early cache recomputation (XFetch) to eliminate thundering herd cache stampedes.	Week 4	20%	25 Hrs (\$1,000)
Milestone 5	Multi-Provider AI Gateway, Rate Limiting & Production Handover Docs Implement sub-500ms automated circuit breaker (Gemini Flash to GPT-4o-mini), inline PII token scrubbing, OWASP LLM01 prompt injection defense, comprehensive integration test suite, and clean architecture runbooks with zero lock-in.	Week 5	20%	25 Hrs (\$1,000)
TOTAL SYSTEMS RELIABILITY ENGAGEMENT (STAFF AUGMENTATION / 20–25 HRS/WK)		30 Days	100%	\$40.00/hr Capped

30-DAY SYSTEMS RELIABILITY ROADMAP

Phase 0: Interactive Architecture Cockpit (Delivered)

\$0.00 (Live Ahead of Bid)

W1: Codebase Discovery & Observability Baseline

25 Hrs (Net Day 7)

W2: PostgreSQL Profiling & Composite Indexing

25 Hrs (Net Day 14)

W3: Ingestion Decoupling & Worker Autoscaling

25 Hrs (Net Day 21)

W4: PgBouncer Pooling & XFetch Stampede Defense

25 Hrs (Net Day 28)

W5: Multi-Provider AI Gateway & Handover Docs

25 Hrs (Net Day 35)

PRODUCTION ARCHITECTURE & RELIABILITY GUARDRAILS

✓ Zero Downtime Migrations: All PostgreSQL indexing deployed CONCURRENTLY with strict lock timeouts.

✓ Database Connection Shield: PgBouncer transaction pooling prevents connection pool exhaustion at peak loads.

✓ Cache Stampede Defense: Redis XFetch probabilistic early refresh eliminates database spikes on expired keys.

✓ Staff Augmentation Hygiene: Clean atomic PRs, daily async standup notes, 100% test pass rate, zero micromanagement.

✓ Cloud Cost Right-Sizing: Optimized query plans & pooling cut AWS RDS and compute bills by up to \$2,450/month.

CALIBRATED HOURLY RATE

\$40.00/hr (20–25 hrs/week). Aligned with client historical average (\$50/hr). Flexible weekly sprint cadence.

CLOUD COST REDUCTION

Halves AWS RDS and serverless compute waste (~\$29,400/yr saved), paying back engineering costs in weeks.

100% CODE OWNERSHIP

All code, PRs, Terraform modules, and Docker configs committed directly to client's Git repositories.

DAILY ASYNC STANDUPS

Daily notes on shipped work, upcoming PRs, blockers, and test status. Zero hand-holding required.

FORMAL AUTHORIZATION & SYSTEMS RELIABILITY ACCEPTANCE

BINDING UPON CONTRACT ACTIVATION VIA UPWORK HOURLY OFFER / MILESTONE SCHEDULE

AUTHORIZED ARCHITECT: BARAKAHSOFT LLC (WYOMING, USA)

AUTHORIZED CLIENT: ENTERPRISE SAAS PLATFORM

Signatory: Shakil Ahmed • Principal Systems Architect & Former Lead Engineer at Legiti

Signatory: Founder / CTO • Authorized SaaS Client Representative

Shakil Ahmed

17 Sep 2026

[Accepted via Upwork Contract Offer / Sign-off]

___ / ___ / 2026

AUTHORIZED ARCHITECT SIGNATURE

DATE

AUTHORIZED CLIENT SIGNATURE

DATE



Shakil Ahmed • Senior Backend Architect & Former Lead Engineer at Legiti (12+ Yrs Exp)

BarakahSoft LLC • Enterprise Systems Engineering & Reliability Architecture

Securiti Certified AI Security & Governance Architect (Cert ID: 14B411BCE-14B411A3D-1451CFE76) • Verified Upwork Partner



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